7. A magnetic moment of 1.73BM will be shown by one among the following (NEET)

a) $TiCl_4$ b) $[CoCl_6]^{4-}$

c) $[Cu(NH_3)_4]^{2+}$ d) $[Ni(CN)_4]^{2-}$

8. Crystal field stabilization energy for high spin d^5 octahedral complex is

a) $-0.6\Delta_0$ b) 0

c) $2(P-\Delta_0)$ d) $2(P+\Delta_0)$

9. In which of the following coordination entities the magnitude of Δ_0 will be maximum?

a) $[Co(CN)_6]^{3-}$ b) $[Co(C_2O_4)_3]^{3-}$

c) $[Co(H_2O)_6]^{3+}$ d) $[Co(NH_3)_6]^{3+}$

10. Which one of the following will give a pair of enantiomorphs?

a) $[Cr(NH_3)_6][Co(CN)_6]$ b) $[Co(en)_2Cl_2]Cl$

c) $[Pt(NH_3)_4][PtCl_4]$ d) $[Co(NH_3)_4Cl_2]NO_2$

11. Which type of isomerism is exhibited by $[Pt(NH_3)_2Cl_2]$?

a) Coordination isomerism b) Linkage isomerism

c) Optical isomerism d) Geometrical isomerism

12. How many geometrical isomers are possible for $[Pt(Py)(NH_3)(Br)(Cl)]$?

a) 3 b) 4 c) 0 d) 15

13. Which one of the following pairs represents linkage isomers?

a) $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$

b) $[Co(NH_3)_5(NO_3)]SO_4$ and $[Co(NH_3)_5(ONO)]$

c) $[Co(NH_3)_4(NCS)_2]Cl$ and $[Co(NH_3)_4(SCN)_2]Cl$

d) both (b) and (c)

14. Which kind of isomerism is possible for a complex $[Co(NH_3)_4Br_2]Cl$?

a) geometrical and ionization b) geometrical and optical

c) optical and ionization d) geometrical only

15. Which one of the following complexes is not expected to exhibit isomerism?

a) $[Ni(NH_3)_4(H_2O)_2]^{2+}$ b) $[Pt(NH_3)_2Cl_2]$

c) $[Co(NH_3)_5SO_4]Cl$ d) $[FeCl_6]^{3-}$

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